

1 Problem Statement

WHY THIS EXISTS

Regression testing of Sous Chef Cloud is performed by hand. Every test case requires a person to log in, click through each step, judge the result against the written expected outcome, record a verdict in QMetry, and raise a Jira bug when something fails.

The scale of that manual work, measured from the live QMetry project:

MEASURE	VALUE	IMPLICATION
Test cases in the project library	2,534	The full regression surface, larger than any team can cover by hand in a release window
Test runs (cycles) on record	414	Each is a repeat pass over overlapping cases — the same clicks, again
Steps per case (observed range)	5–26	Long cases are where attention and consistency fade

Four specific costs follow from doing this manually:

- **Throughput is capped by tester-hours.** Coverage per release is decided by how many people are available, not by which cases carry the most risk.
- **Verdicts vary by person.** Two testers reading the same expected result can record different outcomes, and neither leaves evidence of what they saw.
- **Result entry is manual transcription.** Verdicts are retyped into QMetry and bugs retyped into Jira — effort that produces no testing value and can be mistyped.
- **Repetition is the least rewarding work.** Skilled QA time goes to re-executing known paths instead of exploratory testing, where people outperform automation.

One figure to supply: the average tester-hours for one full manual regression pass is the single number this document cannot derive from the system. It is held by the QA team, not by QMetry, and it is what converts everything below into a cost saving. It is left blank deliberately rather than estimated.

An agent reads test cases from QMetry, translates each plain-English step into a real browser action, executes it against Sous Chef Cloud, judges the resulting screenshot with a vision model, and writes the verdict back — with a live console a tester watches while it runs.

1 · READ	2 · TRANSLATE	3 · EXECUTE	4 · JUDGE	5 · REPORT
Test cases + steps from the QMetry REST API	English step → browser action (AI)	Playwright drives Chromium	Screenshot vs expected (AI vision)	QMetry result, Jira bug, HTML report

What is proven, and how

CAPABILITY	STATUS	EVIDENCE
End-to-end run against the live application	Working	3 of 3 cases passed in ~82 seconds against test.souscheftech.com, with a report generated
QMetry integration (read cases, write results)	Working	Live against cycle SOUSCLOUD-TR-482; case-level result push confirmed
Console: Manual tab plus live execution tape	Working	Browses all 414 runs and 2,534 cases with server-side search; steps load per opened case
Automated regression of the agent itself	374 tests green	Mocked HTTP and browser, so the suite runs offline in seconds
Per-case server targeting (Test / Production / any URL)	Shipped 18 Aug	Validated when saved; warns whenever a case points off the test server
Automatic Jira bug filing	Built, gated off	AUTO_CREATE_BUGS=false until a full-plan run has been verified

Design decisions that matter

- **Credentials never reach the model.** Login and logout are executed by the harness; the AI can request them but never sees a password.
- **The model targets real elements,** not guessed CSS — the page is snapshotted and every action refers to a tagged element on it.
- **Model choices are measured, not assumed.** The evaluator runs gpt-4.1 and the translator gpt-5.4-mini. A comparison harness scored candidates on verdict stability, and one prompt variant was rejected for a 20% flip rate on identical input.
- **A failing case never stops the run.** Each case is isolated; failures are recorded and the run continues.

Known limits, stated plainly

- **Results post per case, not per step.** QMetry's Open API exposes test steps read-only; per-step writes need its separate Automation API.
- **Bug filing is off by default,** and should stay off until a full plan run has been reviewed by a person.
- **Some steps resist single-screenshot judgement** — a 20-item checklist across a scrolling panel is the known example.
- **Tester session state is per machine.** Hand marks and notes do not sync between devices.
- **The QMetry API key needs rotation.** It was exposed in working transcripts and remains valid.

A tester selects a plan, presses Run, and reviews evidence instead of producing it — with the agent covering the repeatable regression surface while people spend their time on judgement and exploration.

HORIZON	OBJECTIVE	GATE TO CLEAR FIRST
Immediate days	Rotate the QMetry API key; confirm the new per-case server row in daily use	Security housekeeping — no dependency
Near weeks	Run a full plan end to end, review every verdict by hand, then enable automatic Jira bug filing	A human-reviewed full-plan run. Auto-filing before that risks noise in the bug tracker
Near weeks	Harden verdict quality: capture the hard evaluation cases plus a should-obviously-fail control, then re-score the evaluator	The current model decision rests on one easy captured step — a known coverage gap
Medium a quarter	Per-step results in QMetry, so a failure points at the step rather than the whole case	Access to QMetry's Automation API (a separate token) — decision pending
Medium a quarter	Scale coverage from a five-case demonstration cycle toward the regression suites that matter most per release	Case text must reference the real UI; poorly-specified cases fail for the wrong reason

What success looks like

- **Regression passes run unattended** — overnight or on demand — with a shareable HTML report carrying a screenshot of every step as evidence.
- **Failures arrive as filed bugs** with steps, screenshots and the model's stated reason, rather than as a tester's recollection.
- **Coverage per release becomes a choice**, not a constraint imposed by available tester-hours.
- **People are pointed at work that needs people**: exploratory testing, new features, and judging the cases the agent flags as uncertain.

The honest boundary: this agent verifies what a test case describes. It does not decide what to test, write good test cases, or replace exploratory testing — and a case whose steps do not match the real UI will fail regardless of how good the agent is. Its value scales with the quality of the test cases it is given.

Figures verified against the live QMetry project and a green test suite on 18 August 2026. Repository github.com/zanieta/QA-Auto. The one deliberately blank figure — manual tester-hours per regression pass — is noted in section 1.